

Quadratic Equation (二次方程)

Consider the equation $ax^2 + bx + c = 0$, where $a \neq 0$

Using Method of completing Squares, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

The discriminant (判別式) $\Delta = (b^2 - 4ac)$

- (i) If $\Delta = 0$, the two roots are identical
- (ii) If Δ is a perfect square, then the roots are rational.
- (iii) If $\Delta > 0$, the two roots are real and unequal, and they are equal to

$$\frac{-b - \sqrt{\Delta}}{2a} \text{ and } \frac{-b + \sqrt{\Delta}}{2a}$$

Example 1: Show that if a, b, c are real and $a > b > c$, the equation $\frac{1}{x-a} + \frac{1}{x-b} + \frac{1}{x-c} = 0$ has two distinct roots lying between c and a .

Example 2: Find all values of x , such that for any value of y , the equation $x^2 + y^2 + z^2 + 2xyz = 1$ has a solution.

Answer: $x = \pm 1$.

Example 3: Show that if the equation $ax^2 + bx + c = 0$ ($a \neq 0$) has rational roots, then not all of a, b and c are odd.

Example 4: Find all real values of a , such that the equation $x^3 - ax^2 + (a^2 - 1)x - a^2 + a = 0$ has three distinct real roots which form an arithmetic progression.

Answer: $a = 0$ or $\frac{6}{7}$

Example 5: (10th IMO 1968 No. 3) Consider the system of equations

$$ax_1^2 + bx_1 + c = x_2$$

$$ax_2^2 + bx_2 + c = x_3$$

.....

$$ax_{n-1}^2 + bx_{n-1} + c = x_n$$

$$ax_n^2 + bx_n + c = x_1$$

with unknowns $x_1, x_2, \dots, x_n$, where a, b, c are real and $a \neq 0$,

Let $\Delta = (b-1)^2 - 4ac$. Prove that for this system

- (a) If $\Delta < 0$, there is no solution.
- (b) If $\Delta = 0$, there is exactly one solution,
- (c) If $\Delta > 0$, there is more than one solution.

Example 6: (15th IMO 1973 No. 3) Let a and b be real numbers for which the equation $x^4 + ax^3 + bx^2 + ax + 1 = 0$ has at least one real solution. For all such pairs (a, b) find the minimum value of $a^2 + b^2$.

Example 7: (3rd IMO 1961 No. 1) Solve the system of equations:

$$x + y + z = a$$

$$x^2 + y^2 + z^2 = b^2$$

$$xy = z^2$$

where a and b are constants. Give the conditions that a and b must satisfy so that x, y, z (the solutions of the system) are distinct positive numbers.

Answer: Minimum value of $a^2 + b^2 = \frac{4}{5}$.

Relationship between coefficients and roots of a quadratic equation

Consider the equation $ax^2 + bx + c = 0$, where $a \neq 0$

- (i) Sum of roots $= -\frac{b}{a}$,
- (ii) Product of roots $= \frac{c}{a}$.

Example 8: If α and β are the roots of the equation $2x^2 - 3x - 4 = 0$, find the quadratic equation whose roots are $\frac{2}{\alpha}$ and $\frac{2}{\beta}$.

Answer: $2x^2 + 3x - 4 = 0$.

Example 9: Given that the sum of the squares of the roots of the equation $x^2 - (m+2)x + m + 7 = 0$ is equal to the square of the length of hypotenuse of a right-angled triangle. The lengths of the sides of the triangle form an arithmetic progression, and their sum is 12. Find the possible value of m .

Answer: $m = -7$ or 5 .

Example 10: If α and β are the roots of the equation $x^2 - x - 1 = 0$, and for any integer n , $f(n) = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n)$, find $f(1)$ and $f(2)$, and show that $f(n+2) = f(n) + f(n+1)$.

Answer: $f(1) = f(2) = 1$.

Example 11: Given that a, b, c are distinct non-zero real numbers, solve the system of equations

$$\begin{aligned}\frac{x}{a^3} - \frac{y}{a^2} + \frac{z}{a} &= 1 \\ \frac{x}{b^3} - \frac{y}{b^2} + \frac{z}{b} &= 1 \\ \frac{x}{c^3} - \frac{y}{c^2} + \frac{z}{c} &= 1\end{aligned}$$

Answer: $x = abc$, $y = ab+bc+ca$, $z = a + b + c$.

Example 12: Prove that if a, b are integers and b is odd, the equation $x^2 - 16ax + 7^b = 0$ has no integral solutions.

Polynomial Equations (多項方程)

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ be a real polynomial in x of degree n .

Next, $f(x) = 0$ is called a real polynomial equation of degree n in x .

Note the following properties

- (i) According to the Fundamental Theorem of Algebra and related Theorems, the equation has n roots, which may be real or complex, and some of them may be double roots, triple roots or roots of multiplicity m .
- (ii) If $a_0 = 0$, then 0 is one of the roots of the equation.
- (iii) If the sum of the coefficients is 0 , then 1 is a root of the equation.
- (iv) If all the coefficients are positive, then the equation has no positive roots.
- (v) If the coefficients are alternatively positive and negative, then the equation has no negative roots.
- (vi) If $a_n = 1$, and all the other coefficients are integers, then there cannot be a non-integral rational root.

Example 13: Prove that the equation $x^{2n} - 2x^{2n-1} + 3x^{2n-2} - \dots + (2n-1)x^2 - 2nx + (2n+1) = 0$ does not have a real root.

Example 14: The coefficients of $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ are all integers. Given that p is odd and q is even, and that both $f(p)$ and $f(q)$ are odd. Show that the equation $f(x) = 0$ does not have integral roots.

Example 15: Given $P(x) = 4x^3 - 2x^2 - 15x + 9$, $Q(x) = 12x^3 + 6x^2 - 7x + 1$, prove that

- (a) Both $P(x) = 0$ and $Q(x) = 0$ have three distinct real roots.
- (b) If α, β are respectively the largest root of $P(x)=0$ and $Q(x)=0$, then $\alpha^2 + 3\beta^2 = 4$.

Theorem on Irrational Roots

Theorem: Given that the coefficients of $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ are all rational numbers. If $a + \sqrt{b}$ is a root of $f(x) = 0$, then so is $a - \sqrt{b}$.

Proof: Consider $g(x) = (x - a - \sqrt{b})(x - a + \sqrt{b}) = x^2 - 2ax + a^2 - b$

Divide $f(x)$ by $g(x)$ gives $f(x) = (x^2 - 2ax + a^2 - b) Q(x) + cx + d$

Put $x = a + \sqrt{b}$, $f(a + \sqrt{b}) = 0$, therefore $c(a + \sqrt{b}) + d = 0$

$$(ca + d) + c\sqrt{b} = 0$$

so $c = 0$, and then $d = 0$

Hence $f(x) = (x^2 - 2ax + a^2 - b) Q(x) = (x - a - \sqrt{b})(x - a + \sqrt{b}) Q(x)$

So, $a - \sqrt{b}$ is also a root of $f(x) = 0$.

Example 16: One of the roots of the equation $x^4 - 22x^2 - 72x + 13 = 0$ is $3 - 2\sqrt{2}$. Find all the roots of the equation.

Answer: $3 + 2\sqrt{2}$, $-3 \pm 2i$.

Complex Roots Occur in Conjugate Pairs (虛根成對原理)

Theorem: Let $f(x)$ be a polynomial with real coefficients. If $a + bi$ ($b \neq 0$) is a root of the equation $f(x) = 0$, then so is $a - bi$.

Proof: $a + bi$ is a root of $f(x) = 0 \Rightarrow (x - a - bi)$ is a factor of $f(x)$.

Now divide $f(x)$ by the quadratic expression $(x - a - bi)(x - a + bi) = (x - a)^2 + b^2$

To get $f(x) = (x - a - bi)(x - a + bi)Q(x) + cx + d$

Since $f(a + bi) = 0$, therefore $c(a + bi) + d = 0 \Rightarrow (ca + d) + cbi = 0$

Equating real and imaginary parts, since $b \neq 0$, therefore $c = 0$, next, $d = 0$.

Hence $f(x) = (x - a - bi)(x - a + bi)Q(x)$, and $a - bi$ is also a root of $f(x) = 0$.

Example 17: One of the roots of the equation $x^6 + x^5 - 3x^4 - 2x^3 + 27x^2 + 25x - 25 = 0$ is $\sqrt{3} + \sqrt{2}i$. Find all the roots of the equation.

Answer: $\pm\sqrt{3} \pm \sqrt{2}i$, $\frac{-1 \pm \sqrt{5}}{2}$

Example 18: Let $x^3 + px^2 + qx + r = 0$ be an equation with real coefficients. If $a + bi$ is a root of $f(x) = 0$, show that $r(a^2 + b^2) = (p + 2a)[(a^2 + b^2)q + 2ar]$

Corollary 1: Every real polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ can be broken into monomial and binomial factor of the form

$$f(x) = a_n(x - x_1) \dots (x - x_m)(x^2 + 2b_1x + c_1) \dots (x^2 + 2b_{\frac{n-m}{2}}x + c_{\frac{n-m}{2}})$$

Example 19: Reduce $x^3 - 1$, $x^3 + 1$, $x^4 - 1$, $x^4 + 1$, $x^6 + 1$, $x^6 - 1$, $x^5 + 1$, $x^5 - 1$, $x^7 - 1$ into monomial and quadratic factors.

Answer: $x^3 - 1 = (x - 1)(x^2 + x + 1)$,

$$x^3 + 1 = (x + 1)(x^2 - x + 1)$$

$$x^4 - 1 = (x - 1)(x + 1)(x^2 + 1)$$

$$x^4 + 1 = (x^2 + 1)^2 - 2x^2 = (x^2 + (\sqrt{2})x + 1)(x^2 - (\sqrt{2})x + 1)$$

$$x^6 + 1 = (x^2 + 1)(x^4 - x^2 + 1) = (x^2 + 1)(x^2 + (\sqrt{3})x + 1)(x^2 - (\sqrt{3})x + 1)$$

$$x^6 - 1 = (x^2 - 1)(x^4 + x^2 + 1) = (x^2 + 1)(x^2 + x + 1)(x^2 - x + 1)$$

$$x^5 + 1 = (x + 1)(x^2 - 2x \cos(\frac{\pi}{5}) + 1)(x^2 - 2x \cos(\frac{3\pi}{5}) + 1)$$

$$x^5 - 1 = (x - 1) \left(x^2 - \left(\frac{1 + \sqrt{5}}{2}\right)x + 1\right) \left(x^2 - \left(\frac{1 - \sqrt{5}}{2}\right)x + 1\right)$$

$$x^7 - 1 = (x - 1) \left(x^2 - 2x \cos\left(\frac{2\pi}{7}\right) + 1\right) \left(x^2 - 2x \cos\left(\frac{4\pi}{7}\right) + 1\right) \left(x^2 - 2x \cos\left(\frac{6\pi}{7}\right) + 1\right)$$

Corollary 2: If a polynomial $f(x)$ is such that the equation $f(x) = 0$ contains only imaginary roots, then $f(x)$ assumes positive values for all values of x .

(Reason: The factors of $f(x)$ are all of the form $(x - a)^2 + b^2$, and they all assume positive values for any value of x .)

Corollary 3: If $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ is a polynomial with rational coefficients and n is odd, then the equation has at least one real root. Further, if the equation has only one real root, then the root must be rational.

(Reason: Complex roots occur in pairs. Further, irrational roots also occur in pairs.)

Example 20: Given $(x) = x^2 + ax + b$, $Q(x) = x^2 + cx + d$. If the equation $P(Q(x)) = Q(P(x))$ does not have a real solution, prove that $b \neq d$.

Descartes Rule of Signs (笛卡兒符號法則)

Theorem: Let $f(x)$ be a polynomial of degree n , with rational coefficients. Consider the equation $f(x) = 0$. The number of positive roots of the equation is equal to the number of sign changes, or less than the number of sign changes by a positive even number. (Note: here a root of multiplicity m is counted as m roots.)

Proof: Suppose the equation $f(x) = 0$ possesses p positive roots, namely $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_p$, q zero roots, and r negative roots namely $-\beta_1, -\beta_2, -\beta_3, \dots, -\beta_r$, together with $2s$ complex roots, $a_1 \pm b_1 i, a_2 \pm b_2 i, a_3 \pm b_3 i, \dots, a_s \pm b_s i$,

Then $f(x) = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3) \dots (x - \alpha_p) \varphi(x)$, where

$$\varphi(x) = x^q (x + \beta_1)(x + \beta_2)(x + \beta_3) \dots (x + \beta_r) (x^2 - 2a_1 x + (a_1^2 + b_1^2)) (x^2 - 2a_2 x + (a_2^2 + b_2^2)) \dots (x^2 - 2a_s x + (a_s^2 + b_s^2))$$

Note that, when expanded, the first term and last term of $\varphi(x)$ are both positive, hence the number of changes in sign is an even number, say $2k$, where $k = 0, 1, 2, \dots$

Multiplying $\varphi(x)$ by $(x - \alpha_1), (x - \alpha_2), (x - \alpha_3), \dots$ and $(x - \alpha_p)$, each time the number of changes of sign is increased by $2k_i + 1$, where $i = 1, 2, 3, \dots, p$.

We may show the above statement by considering the following:

$\varphi(x)$	+	-	-	+	+	-	-	-	+	-	+	-	+
$(x - \alpha_i)$	+	-											
$x\varphi(x)$	+	-	-	+	+	-	-	-	+	-	+	-	+
$(-\alpha_i)\varphi(x)$		-	+	+	-	-	+	+	+	-	+	-	+
$(x - \alpha_i)\varphi(x)$	+	-	+	+	+	-	+	+	+	-	+	-	+

We see that the number of changes in sign is increased by $2k_i + 1$.

$$\text{Hence the total number of changes in sign is } 2k + (2k_1 + 1) + (2k_2 + 1) + (2k_3 + 1) + \dots + (2k_p + 1) = 2k' + p$$

This is the number of changes in the sign of the coefficients in $f(x)$.

The number of positive roots in $f(x) = 0$, is equal to the number of changes in sign (if $k' = 0$), or equal to the number of changes in sign less an even positive number (if $k' \neq 0$).

Corollary 1.: The equation $f(x) = 0$ cannot have more positive roots than there are changes of sign in $f(x)$.

Corollary 2: The equation $f(x) = 0$ cannot have more negative roots than there are changes of sign in $f(-x)$.

Corollary 3: If there are p changes of sign in $f(x)$, and q changes of sign in $f(-x)$, then there are at least $n - p - q$ complex roots.

Example 21: Show that the equation $x^4 + 1 = 0$ has no real roots..

Example 22: Consider $f(x) = x^7 + x^5 + x^4 + 1$. Show that the equation $f(x) = 0$ has 6 complex roots.

Example 23: Consider $x^{10} + x^8 + x^7 - x^6 - x^4 + x^3 + x^2 = 0$. Show that the equation $f(x) = 0$ has at most 8 complex roots.

Example 24: Show that the equation $x^5 + x^3 - x^2 + 2x - 3 = 0$ has one real root and four complex roots.

Example 25: Show that the equation $x^3 - x^2 + 2x + 1 = 0$ does not have any positive root.

Example 26: Two of the roots of the equation $x^3 + 2x^2 - 2x + m = 0$ is equal to 1, find m .

Answer: $m = -1, 0, 3$.

Example 27: Find the least number of complex roots of the equation $x^{3n} - x^{2n} + x^n + x + 1 = 0$.

Relationship between coefficients and roots of a polynomial equation

(Vieta's formulae 韋達公式))

Consider the equation $f(x) = 0$

- (i) Sum of roots $= -\frac{a_{n-1}}{a_n}$
- (ii) Sum of products of roots taking two at a time $= \frac{a_{n-2}}{a_n}$
- (iii) Sum of products of roots taking three at a time $= -\frac{a_{n-3}}{a_n}$
- (iv) ...
- (v) Product of all roots $= (-1)^n \left(\frac{a_0}{a_n}\right)$

Example 28: $P(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + 1$ is a polynomial with non-negative real coefficients. Prove that for any non-negative value of x ,

$$P(x) \geq x^n + {}^nC_1x^{n-1} + {}^nC_2x^{n-2} + \dots + {}^nC_{n-1}x + 1$$

Example 29: Given that the equation $x^3 + ax + b = 0$, where $b \neq 0$, has roots α, β and γ which are all rational numbers. Show that the roots of equation $\alpha x^2 + \beta x + \gamma = 0$ are also rational.

Example 30 Two of the roots of the equation $x^4 - x^3 - 56x^2 + 36x + 720 = 0$ are in the ratio 2:3, and the other two differ by 1, find the roots.

Answer: -6, -4, 5, 6.

Example 31: Show that not all the roots of the equation $x^3 + x^2 - x/2 + 1/6 = 0$ are real.

Example 32: If α, β and γ are the three roots of the equation $x^3 + qx + t = 0$, find the equation whose roots are $(\alpha - \beta)^2, (\beta - \gamma)^2, (\gamma - \alpha)^2$.

Answer: $y^3 + 6qy^2 + 9q^2y + 27t^2 + 4q^3 = 0$.

Example 33: Prove that the roots of the polynomial

$$P(x) = x^n - 5x^{n-1} + 12x^{n-2} - 15x^{n-3} + a_{n-4}x^{n-4} + \dots + a_n$$

cannot all be positive real numbers.

Example 34: (7th IMO, 1965 No. 4): Find all sets of four real numbers x_1, x_2, x_3, x_4 such that the sum of any one and the product of the other three is equal to 2.

Answer: (1, 1, 1, 1), (3, -1, -1, -1). (-1, 3, -1, -1). (-1, -1, 3, -1). (-1, -1, -1, 3).

Example 35: (18th IMO 1976, No. 2): Let $P_1(x) = x^2 - 2$ and $P_j(x) = P_1(P_{j-1}(x))$ for $j = 2, 3, \dots$ Show that for any positive integer n , the roots of the equation $P_n(x) = x$ are real and distinct..

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